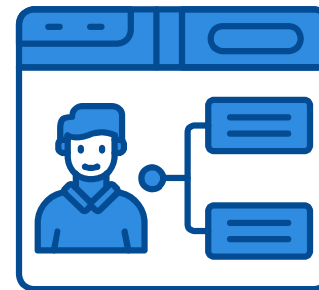


Isolating Signal from Noise in Voxel Sales Calls

A structured analysis of extraction quality and non-safety opportunities



99 Calls

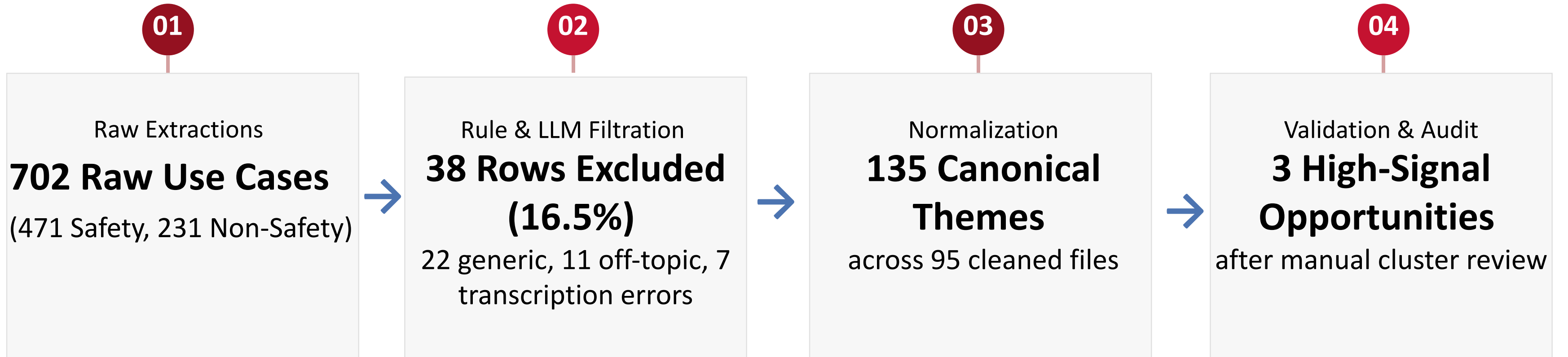


702 Use cases



135 Canonical Themes

Four Stages to Isolate Actionable Product Signal



Pipeline Diagnostic: Strong Quotes, Weak Categorization

Strengths

4.75 / 5

Average evidence support score

4.34 / 5

Average label accuracy score

The pipeline is highly effective at identifying relevant customer quotes from call transcripts.

Weaknesses

207

Quotes appeared in both safety and non-safety buckets within the same file

20

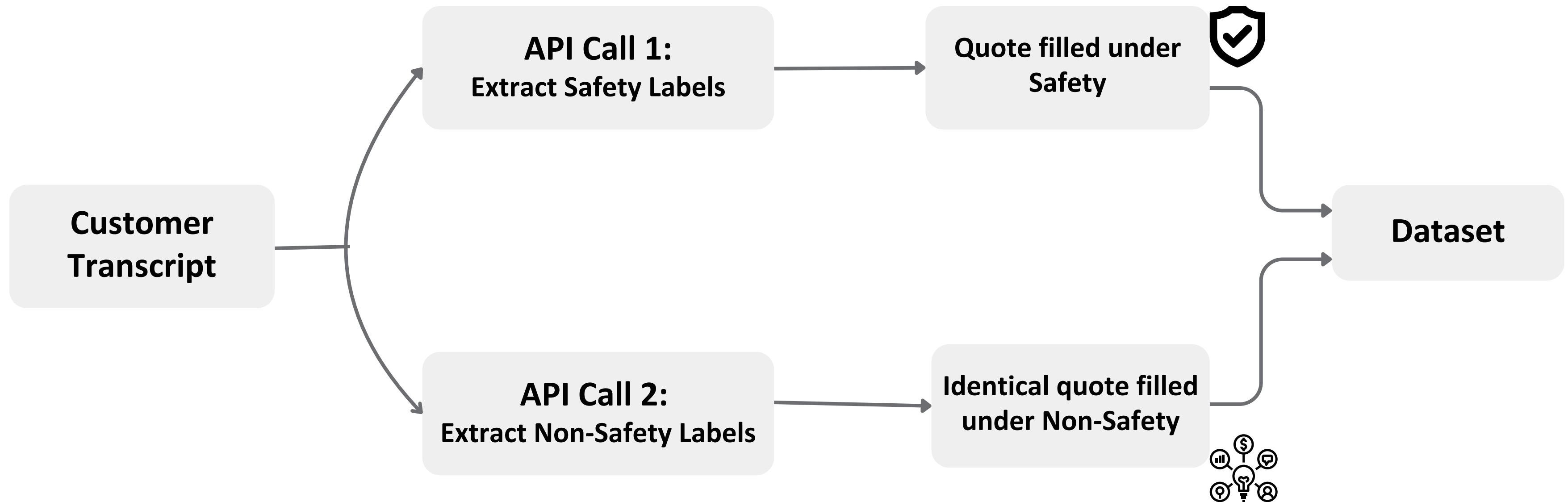
Quotes recycled across multiple labels in a single file

48 of 99

Files affected by cross-bucket contamination

The system struggles with distinct categorization. A single conversation is processed twice, producing duplicate content across buckets.

The Dual-Pass Blindspot Creates Artificial Duplication



DATA PROOF:

The LLM clustering caught 4 of 207 contaminated cases.

Rule-based quote matching found all 207.

Splitting API calls by bucket to stay within token limits made cross-bucket comparison architecturally impossible. This is a design constraint, not a model failure.

High Evidence Scores Do Not Guarantee Valid Opportunities

The Mirage

Action Tracking and Accountability

20 Calls | 4.86/5 Evidence

False Positive

Manual review found all 22 labels described existing Voxel safety workflows, not new opportunities

Disqualified

Existing Product Feature

The Signal

Door Duration Monitoring

11 Calls | 4.40/5 Evidence

Real Opportunity

Manual review confirmed specific operational pain point. Door dwell times impact productivity

Validated

Genuine Product Need

Full Opportunity Ranking with Manual Audit Verdicts

Raw frequency ranking followed by human review of all 10 clusters

#	Canonical Theme	Calls	Evidence	Verdict	Notes
1	Action Tracking and Accountability	20	4.86	DISQUALIFIED	Safety workflow content in all 22 labels
2	Door Duration Monitoring	11	4.4	VALID #1	Genuine ops signal, distinct buyer
3	Dashboard and UI Customization	8	5	VALID #2	Role-based views, enterprise stickiness
4	Safety Analytics and Reporting	8	5	BORDERLINE	Analytics on safety data, core product extension
5	System Integration and Data Export	8	4.62	VALID #3	API/BI integration, platform expansion
6	Product Damage and Shrink Reduction	8	4.22	VALID	New buyer persona, P&L framing
7	Restricted Area and Access Control	7	5	VALID	Loss prevention, after-hours monitoring
8	Warehouse Productivity Optimization	7	4.43	MIXED	One off-topic member (wearables)
9	Report Export and Subscriptions	6	4.75	MIXED	Admin noise mixed in with valid reporting
10	Intervention Impact Tracking	5	5	BORDERLINE	Safety outcome measurement

Three Validated Paths for Non-Safety Expansion

The #1 Signal: Door Duration Monitoring	Opportunity 2: Dashboard and UI Customization	Opportunity 3: System Integration and Data Export	
11 calls 4.40/5 Core Need Track open dock doors during loading cycles to prevent dock leveler damage and improve throughput. Target Buyer Operations and Logistics Managers New Buyer Entry: Operational ROI Breaks Voxel out of the safety-only framing. Sells directly to Ops teams.	8 calls 5/5 Core Need Role-based views, shift-specific dashboards, and gamification for different management layers. Target Buyer Operations Managers and Supervisors Account Retention: Enterprise Stickiness Embeds Voxel into daily routines across multiple management layers. Harder to rip out.	8 calls 4.62/5 Core Need Push Voxel data into existing EHS systems and BI platforms via API or CSV export. Target Buyer EHS Managers and BI/Analytics Teams Ecosystem Play: Platform Expansion Elevates Voxel from a standalone tool to a foundational data feed for enterprise systems.	Honorable Mentions Product Damage & Shrink 8 calls / 4.22 Restricted Area & Access 7 calls / 5 Warehouse Productivity 7 calls / 4.43 Report Export 6 calls / 4.75 Intervention Impact 5 calls / 5

Two Practical Fixes to Improve the Extraction Workflow

Fix 1: Single-Pass Extraction

Problem:

The pipeline runs two separate extraction passes, one for safety and one for non-safety. Because the two passes never see each other, the same quote gets filed in both categories. This produced 207 cross-bucket duplicates across 48 files.

Solution:

Run a single extraction pass that reads the full transcript and forces the model to commit to one category per quote.

Route A goes to safety, Route B to non-safety, Route C is discarded. The model is never allowed to file the same evidence in two places.

Expected impact: eliminates cross-bucket contamination at the source

Fix 2: Require 2-3 Quote Turns Per Use Case

Problem: The pipeline accepts a single fragment quote as sufficient evidence for a use case. Door Duration Monitoring appeared in 12% of calls but most evidence is a one-line mention. The pipeline knows something was discussed but loses the why.

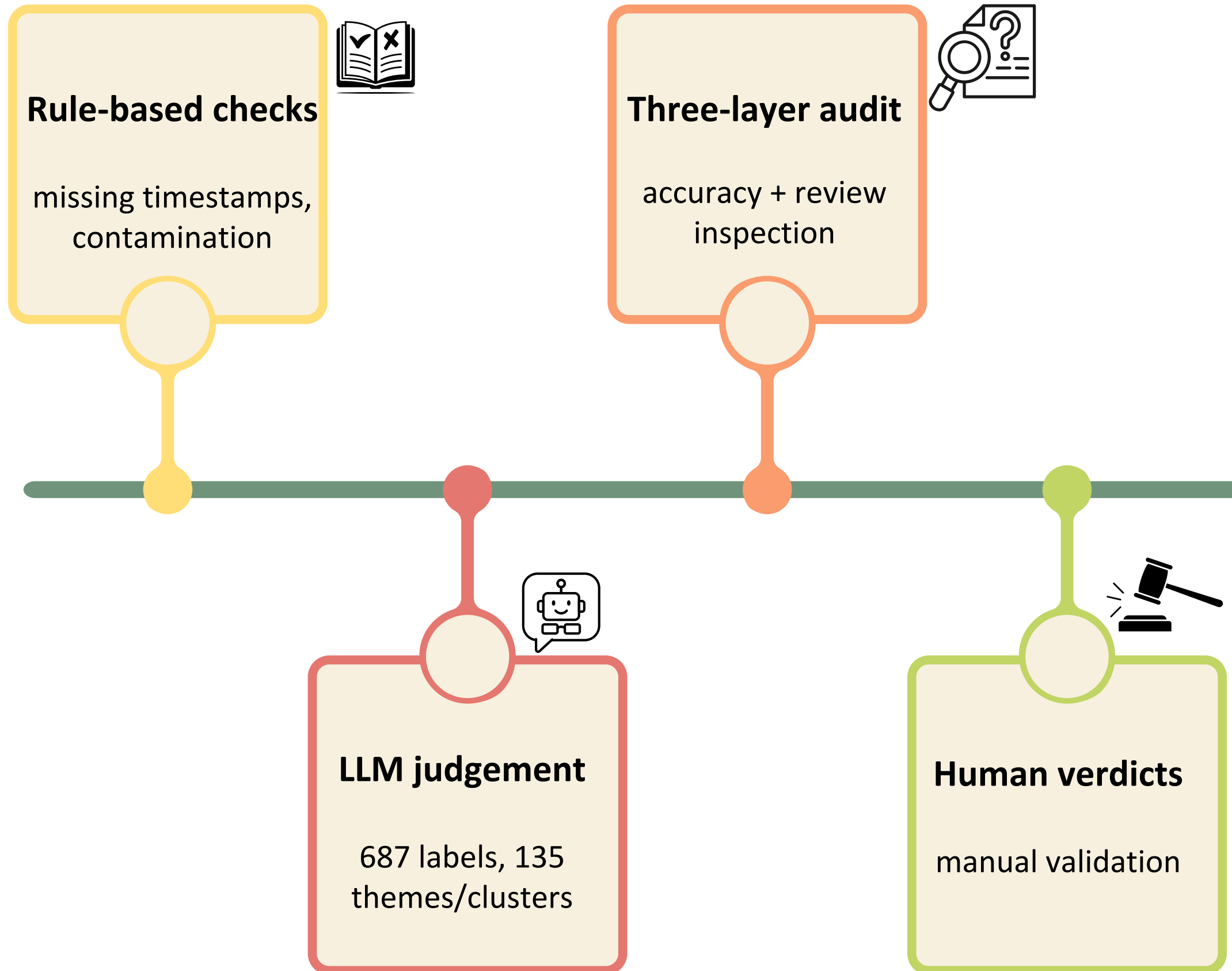
Solution:

Require 2 to 3 consecutive conversational turns to validate a use case extraction.

A customer articulating a problem, asking a question, and confirming a solution is a use case. A customer mentioning a feature name in a list is not a use case.

Expected impact: reduces manual audit burden and raises signal confidence

Approach and Limitations



Limitations to flag

- 1) Small ground truth sample**
8 cases only,
→ recommended 20-30
- 2) Singleton rate 34%**
Labels left ungrouped
→ opportunity underestimated
- 3) One-pass clustering**
Token-limit batching → cross
bucket contamination risk
- 4) Subjective audit**
Human judgment → possible
reviewer variation